



Telegram bots

Telegram is the latest to join the trend of setting bots up to deliver news or trending content into the app itself <http://dgit.in/TeleBot>

Wheeler FCC chairman

Tom Wheeler the FCC chairman rejected the new broadband price cap and last mile unbundling proposal to treat them as common carriers. <http://dgit.in/ChairFCC>

From the labs

All hail, Passive Wi-Fi!

A true breakthrough in wireless communication

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Since time immemorial mankind has strived towards convenience and comfort. Our ability to introduce convenience in everything we do is only outmatched by our ingenuity to make our lives more comfortable through it. And thanks to some student geniuses over at the University of Washington, everyone's increasingly indispensable digital comfort – the gateway to our digital addiction – i.e. Wi-Fi networks are going to become a whole lot efficient. This in turn will accelerate the practical implementation of a whole new era of digital interconnectedness that we've all been waiting for – IoT.

As a technological breakthrough, Passive Wi-Fi has some astonishing claims to its name. Claims that will have networking brands beelining at the gates of University of Washington and Jeeva Wireless – a startup that plans to commercialize the technology – eager to get a piece of the impending wireless revolution.

But what is Passive Wi-Fi?

It's a research project conducted by two PhD students (Bryce Kellogg and Vamsi Talla) in the Electrical Engineering department at University of Washington, a study that conclusively aims to reduce the power draw in wireless communications by several thousand times, a crucial breakthrough to extend Wi-Fi in low-powered sensors and long-term embed-

ding scenarios. Just how power-efficient is Passive Wi-Fi? It claims to be 10,000 times more energy efficient than conventional Wi-Fi, and up to 1,000 times less power consuming than "low powered" Bluetooth and ZigBee networks. Yeah, let that sink in for a moment.

How is it different than conventional Wi-Fi?

The answer to this question can't be appreciated fully unless we take a closer look at the radio antenna used in conventional Wi-Fi routers and beacons. These Wi-Fi radios have two critical components – a digital baseband and analog RF, where the baseband is responsible for the digital information sent or received via analog radio waves. The problem arises

